

Auxiliary Electric Motors in European Battery Electric Vehicles (2024-2026 Models): A Technical Analysis

Report Date: 2026-02-17

Executive Summary

This report provides a detailed technical analysis of non-drivetrain auxiliary electric motors in Battery Electric Vehicles (BEVs) projected for the European market between 2024 and 2026. The research focuses on three distinct motor categories: small stepper motors with a power rating under 50 watts, medium DC motors rated between 50 and 500 watts, and medium stepper motors with a power rating of 50 to 200 watts. The proliferation of sophisticated comfort, safety, and efficiency systems in modern BEVs has led to a significant increase in the number and complexity of these auxiliary motors. They perform critical functions ranging from cabin climate control and adaptive lighting to thermal management of the battery and power electronics. This analysis identifies the specific vehicle systems employing these motors, their precise locations, their functional roles, and their typical technical specifications, including power ratings. The findings are substantiated with data from automotive industry documentation, technical papers, and manufacturer specifications, providing a comprehensive overview of the current and near-future landscape of auxiliary motor applications in European and Asian BEVs sold in Europe. Key applications for small stepper motors include HVAC blend door actuators and adaptive headlamp systems, where their precision is paramount. Medium DC motors are predominantly found in essential thermal management systems, such as electric water pumps for coolant circulation. The category of medium stepper motors, while specified for this research, appears less prevalent in the identified common auxiliary applications, which tend to favor either the high precision of smaller steppers or the continuous power of DC motors.

Introduction

The transition to battery electric vehicles represents a fundamental shift in automotive engineering, extending far beyond the replacement of the internal combustion engine with an electric drivetrain. A critical, yet often overlooked, aspect of this evolution is the extensive electrification of auxiliary vehicle systems. In a BEV, every function that once relied on mechanical or vacuum power derived from a running engine must now be driven by an electric motor. This has led to the integration of a diverse array of small and medium-sized electric motors responsible for functions essential to vehicle safety, passenger comfort, and operational efficiency. The energy consumption of these auxiliary systems has a direct and significant impact on the vehicle's overall driving range, making the efficiency and design of their motors a crucial consideration for automotive manufacturers.

This report aims to deliver a focused investigation into these auxiliary electric motors, specifically within the context of European and Asian BEV models anticipated for sale in the European market for the 2024-2026 model years. The analysis is structured around three specific classifications of motors: small stepper motors (<50W), medium DC motors (50-500W), and medium stepper motors (50-200W). For each category, this report will detail the specific applications, in-vehicle locations, technical characteristics, and typical power ratings. By synthesizing information from technical reviews, component specifications, and industry publications, this document provides an authoritative overview for stakeholders seeking to understand the component-level architecture of modern electric vehicles.

Small Stepper Motors (<50 Watts)

Small stepper motors, characterized by power ratings typically below 50 watts, are integral to a multitude of precision control systems in modern BEVs. Their defining characteristic is the ability to rotate in discrete, fixed angular increments, or “steps,” which allows for highly accurate positioning without the need for complex and costly feedback sensors. This capability, known as **open-loop control**, makes them exceptionally reliable and cost-effective for applications where precise movement is more critical than high speed or torque [3, 7]. In the context of BEVs, where energy conservation is paramount to maximizing driving range, the efficiency and low power draw of these motors are significant advantages. They are commonly employed in systems that require the exact positioning of components, such as flaps, valves, and lenses. Their integration is often managed through dedicated control units that communicate over vehicle data networks like the Local Interconnect Network (LIN) bus, enabling sophisticated, coordinated operation [6, 7].

HVAC System Actuators

One of the most prevalent applications for small stepper motors is within the Heating, Ventilation, and Air Conditioning (HVAC) system [1, 3, 4]. These motors function as actuators, driving the various doors and flaps that control air temperature, distribution, and source. Located deep within the main HVAC module, which is typically situated behind the vehicle’s dashboard and firewall, these actuators are critical for maintaining passenger comfort. A key function is the operation of the **blend door**, which regulates cabin temperature by precisely mixing hot air from the heater core with chilled air from the evaporator [3, 4]. A stepper motor allows the vehicle’s climate control unit to position this door with great accuracy, achieving the exact temperature selected by the occupants. According to a 2019 review published by Transstellar Journals, the use of stepper motors in this role can reduce the energy consumption of the actuator system by as much as 20.15% compared to traditional DC motors, a vital saving in a BEV [3, 4].

Beyond temperature regulation, stepper motors also control the **mode doors**, which direct the flow of air to different cabin vents, such as the windscreen defroster, dashboard face vents, or floor-level footwell vents. They also operate the **recirculation flap**, which switches the system’s air intake between fresh outside air and recirculated cabin air. The precise, repeatable positioning offered by stepper motors ensures consistent performance and allows for complex climate control strategies. While specific power ratings are not always published, the function demands low torque and speed, placing these motors comfortably within the sub-50-watt category. Their design must also account for the demanding thermal environment within the dashboard, where operational temperatures can exceed 78°C, impacting motor longevity [3]. The use of these actuators is standard across the industry, found in BEVs from manufacturers like Mercedes-Benz and others, where efficient and silent operation is a key element of the premium passenger experience [2].

Adaptive and Automatic Headlamp Systems

Another critical safety application for small stepper motors is in advanced headlamp systems. Modern BEVs are frequently equipped with automatic headlamp leveling and Adaptive Front-lighting Systems (AFS), both of which rely on stepper motors for their precise and rapid adjustments [6, 7]. These motors are housed directly within the headlamp assemblies [7]. European regulations mandate automatic headlamp leveling for certain types of bright headlights, such as HID xenon and high-power LEDs, to prevent dazzling oncoming drivers [6, 7]. The system uses vehicle level sensors, often mounted on the front and rear axles, to detect changes in the vehicle’s pitch due to passenger load, cargo, or dynamic events like acceleration and braking. A control unit processes this sensor data and commands the stepper motors to make minute vertical adjustments to the headlamp angle, keeping the light beam parallel to the road surface [6, 7].

In more advanced Adaptive Front-lighting Systems, stepper motors provide horizontal, or swivel, adjustments. As detailed in an analysis by EE Times, AFS uses real-time data on steering angle and vehicle speed from the CAN bus to pivot the headlamps into a turn, illuminating the road ahead rather than the verge [7]. This significantly improves driver visibility and reaction time when cornering. Each headlamp assembly in an AFS typically contains two separate stepper motors: one for vertical leveling and another for horizontal swiveling [7]. These motors are engineered for quiet, low-resonance operation, often through micro-stepping control techniques. HELLA, a major automotive supplier, has developed “Intelligent Stepper Motors” (ISM) that integrate the motor and power electronics into a single mechatronic unit with a LIN-bus interface for diagnostics and control [6]. The power consumption for these motors is very low; specifications for similar DC motors in this application suggest a current draw of around 0.5 amps at 12 volts, equating to just 6 watts [8]. Stepper motors in this application operate in a similar power range, with typical torque requirements between 0.5 and 1 Nm and low rotational speeds of 2 to 10 RPM, ensuring smooth and controlled beam movement [8].

Other Small Actuator Applications

Small stepper motors and functionally similar small DC motors are also utilized in a variety of other low-power actuator roles throughout the vehicle. One such application in BEVs is the **charging port door actuator** [16, 18]. This motorized mechanism is responsible for locking and unlocking the flap that covers the vehicle’s charging port. Integrated with the central locking system, this actuator ensures the port is secure when the vehicle is locked and accessible for charging when unlocked [17]. For example, genuine OEM replacement actuators are listed for various BMW electric models, including the i3 and iX series [16, 18]. While the specific motor type is not always detailed, the function requires a simple, low-power action, making it suitable for either a small stepper or a basic DC motor, with power consumption well below the 50-watt threshold.

Similarly, **active grille shutters**, which are increasingly common on BEVs to improve aerodynamic efficiency, employ small actuators. These systems feature motorized louvers in the front grille that automatically close at higher speeds to reduce aerodynamic drag and open when cooling is needed for the battery or power electronics. The actuators that control these louvers are often described as BLDC stepper actuators or simply small motor-driven actuators [11]. They must be robust, compact, and watertight to withstand their exposed location at the front of the vehicle. By precisely controlling airflow, they contribute directly to optimizing the vehicle’s energy consumption and extending its driving range. The power required for this slow, deliberate movement is minimal, placing these motors firmly in the small, sub-50-watt category.

Medium DC Motors (50 - 500 Watts)

Medium-power Direct Current (DC) motors, particularly of the brushless (BLDC) variety, are workhorses in BEV auxiliary systems that require continuous operation and higher power output than small actuators. This category, defined for this report as motors with a power rating between 50 and 500 watts, is essential for fluid-pumping applications, most notably within the vehicle’s thermal management systems. Unlike stepper motors, which excel at precise positioning, DC motors are designed for sustained rotational speed and torque, making them ideal for driving pumps and fans. In BEVs, the absence of a combustion engine necessitates a fully electrified approach to managing heat generated by the high-voltage battery, power electronics, and the electric drive motor itself, as well as providing heat for the passenger cabin. Efficient and reliable thermal management is not just a matter of comfort but is fundamental to battery health, performance, and longevity.

Electric Water Pumps for Thermal Management

The most significant application for medium DC motors in BEVs is driving the **electric water pumps** that circulate liquid coolant throughout the vehicle's various thermal management loops. A modern BEV may have multiple, independent cooling circuits. A primary loop is dedicated to maintaining the high-voltage battery pack within its optimal temperature range (typically 20-40°C). Another loop may manage the temperature of the power inverter and the main drive motor. A third circuit, often integrated with a heat pump system, is responsible for cabin heating and cooling. Each of these circuits requires at least one electric water pump to ensure the continuous flow of coolant. These pumps are typically located in the engine bay or frunk area, near the components they service, or integrated into larger thermal management modules.

The power consumption of these pumps is a critical design parameter and varies based on the specific requirements of the cooling loop, such as the required flow rate and pressure. Analysis of electric pump specifications indicates that power ratings commonly fall within the 50-500 watt range. For instance, data from Explorist.life on 12V systems shows that a pump drawing 7.5 amps consumes 90 watts ($\text{Power} = 7.5\text{A} \times 12\text{V}$) [22]. Other sources, such as EnergyBot, cite an average power consumption of 150 watts for a general-purpose water pump [21]. These figures align with the demands of automotive coolant circulation. The total energy usage depends on the pump's run time, which is managed by the vehicle's central thermal management controller. The pump may run continuously while the vehicle is in operation or cycle on and off as needed to maintain target temperatures, especially during fast charging, which generates significant heat. The use of efficient brushless DC motors is standard in this application due to their long lifespan, high reliability, and precise speed control capabilities, which allow the system to adjust coolant flow dynamically based on real-time thermal loads, thereby optimizing energy consumption.

Medium Stepper Motors (50 - 200 Watts)

The category of medium stepper motors, with power ratings between 50 and 200 watts, represents a more specialized segment within automotive applications. While small stepper motors are ubiquitous for low-power precision control and DC motors are favored for continuous-duty tasks like pumping, applications requiring the specific combination of high precision and moderate torque offered by a medium-power stepper motor are less common in the auxiliary systems examined for this report. The primary advantages of stepper motors—precise angular control in an open-loop configuration—are most pronounced in smaller-scale applications where the load is predictable and relatively light. As power and torque requirements increase, the complexity and cost of the driving electronics for stepper motors also rise, and brushless DC motors with feedback sensors (closed-loop control) often become a more practical and efficient solution for continuous or high-load tasks.

Potential and Niche Applications

Despite their less frequent use, there are theoretical and niche areas where medium stepper motors could be applied in advanced BEV systems. One potential area could be in sophisticated, high-load active aerodynamic systems. While current active grille shutters use low-power actuators, future systems with larger, faster-moving aerodynamic elements, such as active spoilers or diffusers that require precise positioning for optimal downforce or drag reduction, could potentially leverage the capabilities of a medium-power stepper motor. Another hypothetical application could be within advanced steering or suspension systems, such as steer-by-wire or active roll stabilization, where precise and forceful actuation is needed. However, these critical safety systems typically employ highly redundant and robust brushless DC motors with closed-loop feedback for maximum reliability and fail-safe operation.

Based on the available research data covering common auxiliary systems in 2024-2026 model year BEVs, there is no widespread, clearly identified application that consistently utilizes stepper motors in the 50-200 watt power range. The majority of non-drivetrain motor applications appear to be well-served by either the sub-50-watt stepper motors for precision positioning tasks or the 50-500 watt DC/BLDC motors for continuous fluid pumping and fan operation. Therefore, this category remains largely specialized, and its deployment in the European BEV market within the specified timeframe is not prominently documented in the reviewed sources. Further investigation into highly specialized or proprietary systems from individual manufacturers would be required to identify specific use cases.

Conclusion

The auxiliary systems of modern battery electric vehicles are a complex ecosystem of electrified components, with small and medium electric motors playing a foundational role. This analysis, focused on European and Asian BEVs for the 2024-2026 European market, confirms the distinct and critical roles of different motor types based on their power and functional characteristics.

Small stepper motors, with power ratings under 50 watts, are the definitive choice for applications demanding high precision and open-loop control. Their deployment in HVAC blend door actuators and adaptive headlamp systems is standard practice, enabling sophisticated climate control and advanced safety lighting features while contributing to overall vehicle energy efficiency. The move towards integrated mechatronic units, such as intelligent stepper motors with built-in electronics and network interfaces, highlights a trend towards smarter, more distributed control architectures.

Medium DC motors, particularly brushless variants in the 50 to 500-watt range, are indispensable for the continuous-duty work of fluid circulation. Their primary role in driving electric water pumps for the thermal management of the battery, powertrain, and cabin is fundamental to the performance, longevity, and safety of the entire vehicle. The efficiency and controllability of these motors are paramount, as their energy consumption directly impacts the vehicle's effective driving range.

The category of medium stepper motors (50-200W) appears to be a niche segment with no broadly identified application in the common auxiliary systems reviewed. The industry preference leans towards smaller steppers for precision tasks and more powerful DC motors for continuous operation, leaving the middle ground for specialized or future applications that have yet to become mainstream.

In summary, the selection of an auxiliary motor in a BEV is a carefully considered engineering decision, balancing precision, power, efficiency, and cost. As BEV technology continues to advance, the role and sophistication of these non-drivetrain motors will only grow, further shaping the performance and character of the next generation of electric vehicles.

References

1. [HVAC Heater Blend Door Actuator - Duallane Truck Parts](https://www.duallane.com/shop-parts/hvac/hvac-blowers-controls/hvac-heater-blend-door-actuator) (<https://www.duallane.com/shop-parts/hvac/hvac-blowers-controls/hvac-heater-blend-door-actuator>)
2. [000906920864 - Stepper Motor 2019-2025 Mercedes-Benz - Mercedes-Benz USA](https://parts.mbusa.com/oem-parts/mercedes-benz-hvac-blend-door-actuator-9069208) (<https://parts.mbusa.com/oem-parts/mercedes-benz-hvac-blend-door-actuator-9069208>)
3. [\(PDF\) STEPPER MOTOR ACTUATOR FOR HVAC BLEND DOOR -A REVIEW - Academia.edu](https://www.academia.edu/40175828/STEPPER_MOTOR_ACTUATOR_FOR_HVAC_BLEND_DOOR_A_REVIEW) (https://www.academia.edu/40175828/STEPPER_MOTOR_ACTUATOR_FOR_HVAC_BLEND_DOOR_A_REVIEW)

4. [Stepper Motor Actuator for HVAC Blend Door - A Review by Transtellar Publications - Issuu - Transtellar Publications](https://issuu.com/tjprc/docs/34.ijmperdaug201934) (https://issuu.com/tjprc/docs/34.ijmperdaug201934)
5. [13192013 by OES | Blend Door Actuator \(Stepper Motor\) - eSaabParts](https://farm1.staticflickr.com/819/40121813855_c177e92b6c_k.jpg) (https://farm1.staticflickr.com/819/40121813855_c177e92b6c_k.jpg)
6. [Headlamp Levelling Systems - HELLA](https://www.hella.com/techworld/us/technical/automotive-lighting/headlamp-levelling-system/) (https://www.hella.com/techworld/us/technical/automotive-lighting/headlamp-levelling-system/)
7. [Stepper motors assist adaptive headlights - EE Times](https://www.eetimes.com/stepper-motors-assist-adaptive-headlights/) (https://www.eetimes.com/stepper-motors-assist-adaptive-headlights/)
8. [headlamp leveling motors - Alibaba.com](https://www.alibaba.com/showroom/headlamp-leveling-motors.html) (https://www.alibaba.com/showroom/headlamp-leveling-motors.html)
9. [Automotive 2-Channel Stepper Motor Driver for Dynamic Headlight Reference Design - Texas Instruments](https://www.ti.com/tool/TIDA-020026) (https://www.ti.com/tool/TIDA-020026)
10. [Stepper motors assist adaptive headlights - EDN](https://www.edn.com/stepper-motors-assist-adaptive-headlights/) (https://www.edn.com/stepper-motors-assist-adaptive-headlights/)
11. [Actuator-Motor,5877R1006 5877R1007 BLDC Stepper Shutter Actuator Motor Compatible with 2016-2021 Cherokee,2019-2021 1500,2021-2022 Escape - Amazon.com](https://ae01.alicdn.com/kf/S1981e68a0af94b139f56852a80617290B.jpg_960x960.jpg) (https://ae01.alicdn.com/kf/S1981e68a0af94b139f56852a80617290B.jpg_960x960.jpg)
12. [Pleoss Idle Speed Control Valve, Replacement for King Kong Idle Motor 90380-10526 90325864 - Amazon.com](https://i.ebayimg.com/images/g/NXgAAOSwsrpne1CV/s-l1200.jpg) (https://i.ebayimg.com/images/g/NXgAAOSwsrpne1CV/s-l1200.jpg)
13. [JawGrew Automotive Throttle Stepper Motor, Adjustable Idle Speed Control Valve, Compatible with King Kong Idle Motor - Amazon.com](https://i.pinimg.com/originals/17/e7/b9/17e7b9f7a3a5bfa2f24f4a7cc18a5f9b.jpg) (https://i.pinimg.com/originals/17/e7/b9/17e7b9f7a3a5bfa2f24f4a7cc18a5f9b.jpg)
14. [gunroil Automotive Throttle Stepper Motor, Modified Accessories, Compatible with King Kong Idle Motor - Amazon.com](https://i.pinimg.com/originals/6a/85/cb/6a85cb44c062b80b17118b34b8136ace.jpg) (https://i.pinimg.com/originals/6a/85/cb/6a85cb44c062b80b17118b34b8136ace.jpg)
15. [90380-10526 90325864 Throttle Stepper Motor, Stable Operation - Amazon.com](https://www.amazon.com/90380-10526-90325864-Throttle-Operation/dp/B0DP4RPRRH) (https://www.amazon.com/90380-10526-90325864-Throttle-Operation/dp/B0DP4RPRRH)
16. [Genuine OEM Drive Motor Battery Pack Charging Port Door Release Actuator For BMW - eBay](https://i.ebayimg.com/images/g/pawAAOSwoo5mfqW1/s-l1200.jpg) (https://i.ebayimg.com/images/g/pawAAOSwoo5mfqW1/s-l1200.jpg)
17. [Control, central locking system - Smoothbev](https://smoothbev.com/products/control-central-locking-system/) (https://smoothbev.com/products/control-central-locking-system/)
18. [Genuine OEM Drive Motor Battery Pack Charging Port Door Release Actuator For BMW I01 i3 i3s 2014-2020 - Amazon.com](https://www.amazon.com/Genuine-Battery-Charging-Actuator-2014-2020/dp/B0CL72N892) (https://www.amazon.com/Genuine-Battery-Charging-Actuator-2014-2020/dp/B0CL72N892)
19. [Charging Port Release Actuator - Smoothbev](https://smoothbev.com/products/charging-port-release-actuator-3) (https://smoothbev.com/products/charging-port-release-actuator-3)
20. [Actuator, charge port - Smoothbev](https://smoothbev.com/products/actuator-charge-port) (https://smoothbev.com/products/actuator-charge-port)
21. [Water Pump Energy Calculator: Watts and kWh - EnergyBot](https://www.energybot.com/energy-usage/water-pump.html) (https://www.energybot.com/energy-usage/water-pump.html)
22. [Calculating Water Pump Power Consumption - Explorist.life](https://explorist.life/water-pump-how-much-for-a-mobile-marine-or-off-grid-electrical-system/) (https://explorist.life/water-pump-how-much-for-a-mobile-marine-or-off-grid-electrical-system/)
23. [Submersible pump power consumption - Solar-Electric.com Forum](https://forum.solar-electric.com/discussion/357504/submersible-pump-power-consumption) (https://forum.solar-electric.com/discussion/357504/submersible-pump-power-consumption)
24. [r/SolarDIY on Reddit: Calculating power usage of water pump - Reddit](https://www.reddit.com/r/SolarDIY/comments/pxb31p/calculating_power_usage_of_water_pump/) (https://www.reddit.com/r/SolarDIY/comments/pxb31p/calculating_power_usage_of_water_pump/)

25. [How Many Watts Does a Well Pump Use | Nature's Generator - Nature's Generator](https://natures-generator.com/blogs/news/how-many-watts-does-a-well-pump-use) (<https://natures-generator.com/blogs/news/how-many-watts-does-a-well-pump-use>)